t3-w09 — condition C1

Need for statistic — Need for statistics: purposes, and the collection and tabulation of data

> The objective plus the lesson plan's design: position, method labels, materials, entry behaviour and assessment.

INPUT LAYERS

Input layer	What it carries	Channel	In this condition
Scope	what is to be taught, and to whom	prompt	**present**
Prior knowledge	what the learner already knew coming in	prompt	**present**
Materials	what the lesson was represented with	prompt	**present**
Position	where this objective sat in the sequence	prompt	**present**
Method	the shape of the teaching, as event labels	prompt	**present**
Enacted content	what was actually explained and worked through	prompt	absent
Set work	what the learners were asked to do in class	prompt	absent
Assessment	the questions the lesson was checked with	prompt	**present**
Figures	the drawings the lesson used	tool	absent

WHAT THE MODEL IS SENT

Contract `ComposeWriteObjectiveSection`, declaring: `artefact_spec, lesson_context, unit_context`.

Lesson context — shared by every section

`subject`

Mathematics

`grade_level`

JSS1

`topic`

Need for statistic

`subtopic`

Need for statistics: purposes, and the collection and tabulation of data

`term`

Third Term

`week`

9

`lesson_objectives`

• explain the purpose of statistic
• analyse and interpret statistical data presented in tables

`entry_behaviour`

• Students are familiar with counting and recording information.

`instructional_materials`

• classroom register, bottle tops, pencils or counters.

Objective 1 — explain the purpose of statistic

`objective_text`

explain the purpose of statistic

`current_step`

1

`instructional_events`

• present_content

`assessment_points`

• (a) Define statistic

Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `student_tasks`, `activities`, `source_evidence`, `source_figures`, `objective_learner_state` — carried by the plan or the unit, not sent under this condition.

Objective 2 — analyse and interpret statistical data presented in tables

`objective_text`

analyse and interpret statistical data presented in tables

`current_step`

2

`assessment_points`

• (b) State three methods of collecting data.
• (c) Prepare a frequency table for the following data: 1, 1, 3, 4, 4, 3, 2, 2, 3, 3, 4, 1, 2.

Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `instructional_events`, `student_tasks`, `activities`, `source_evidence`, `source_figures`, `objective_learner_state` — carried by the plan or the unit, not sent under this condition.

WHAT THE MODEL PRODUCED

Objective 1

Why We Use Statistics

> EXPLANATION
> Statistics is the science of collecting, organising, and interpreting data so that we can make sense of the world around us. It helps us answer questions such as:
> - How many students in our school like chocolate?
> - Which fruit is the most popular in the market?
> - What is the average height of our class?
> By turning raw numbers into tables, charts, and summaries, statistics lets us see patterns, compare groups, and make informed decisions.

> WORKED_EXAMPLE
> Example: Imagine you counted the number of bottle-top colours in the classroom and got the following data:
>
> | Colour | Number of tops |
> |---|---|
> | Red | 12 |
> | Blue | 8 |
> | Green | 5 |
> | Yellow | 5 |
>
> From this table you can quickly see that red tops are the most common, and that the total number of tops is 30. This information could help you decide how many red caps to order for a school project.

1. You asked your classmates how many pets they have and recorded the answers: | Number of pets | Number of students | |---|---| | 0 | 4 | | 1 | 6 | | 2 | 3 | | 3 | 2 | What does this table tell you about the most common number of pets among the students?

answer: The most common number of pets is 1, because the highest frequency (6 students) is in that row.

2. A survey of favourite sports gave the following results: | Sport | Votes | |---|---| | Football| 15 | | Basketball| 9 | | Athletics| 6 | | Swimming| 4 | Identify one purpose of collecting this data.

answer: To find out which sport is most popular so the school can plan facilities or events accordingly.

3. You recorded the number of books read by each student in a week: | Books read | Students | |---|---| | 0 | 2 | | 1 | 5 | | 2 | 7 | | 3 | 3 | | 4 | 1 | What is the total number of books read by the whole class?

answer: Total = (0×2)+(1×5)+(2×7)+(3×3)+(4×1) = 0+5+14+9+4 = 32 books.

Objective 2

Understanding and Interpreting Frequency Tables

> EXPLANATION
> Statistics helps us make sense of large amounts of information. One common way to organise data is a **frequency table**, which shows how many times each value occurs. By looking at a frequency table you can quickly see which values are most common, spot patterns, and compare groups.
>
> A frequency table has two columns:
> 1. **Value (or class)** – the distinct items or numbers that appear in the data.
> 2. **Frequency** – the count of how many times each value occurs.
>
> Reading a frequency table is straightforward: start at the top, note the value, then read its frequency. The total of all frequencies should equal the number of data points you started with.

> WORKED_EXAMPLE
> **Worked Example:**
>
> Data set: 2, 5, 3, 2, 4, 5, 5, 3, 2
>
> 1. List the distinct values in order: 2, 3, 4, 5.
> 2. Count how many times each appears:
> - 2 appears 3 times
> - 3 appears 2 times
> - 4 appears 1 time
> - 5 appears 3 times
> 3. Write the frequency table:
>
> | Value | Frequency |
> |---|---|
> | 2 | 3 |
> | 3 | 2 |
> | 4 | 1 |
> | 5 | 3 |
>
> Check: 3 + 2 + 1 + 3 = 9, which matches the total number of data points.

1. (b) List three different methods you could use to collect data about your classmates' favourite school subjects.
answer: 1. Survey questionnaire 2. Interview 3. Observation

2. (c) Prepare a frequency table for the following data: 1, 1, 3, 4, 4, 3, 2, 2, 3, 3, 4, 1, 2.
answer: | Value | Frequency | |---|---| | 1 | 3 | | 2 | 3 | | 3 | 4 | | 4 | 3 |

LAYER BY LAYER: WHAT WAS GROUNDED AND WHAT WAS NOT

Output layer	Made	Written from	Verdict
Title the reader-facing framing of the section	2	Scope	**grounded**
Explanation the principle, rule or relationship to review	2	Method written from event labels — the shape of the teaching without its substance	**substituted**
Worked example a problem with its solution worked through	2	nothing in the input	**unsourced**
Figure a drawing the learner reads from	0	nothing in the input	not produced

Practice what the learner does	5	Assessment written from the teacher's evaluation questions instead of her set work	**substituted**
Answer what the learner checks against	5	Assessment checked against the evaluation answers rather than the taught method	**substituted**